

Self-Learning: Distributed Transactions, TCL & Performance Tuning DBMS

> **Definition:** Distributed Transactions** execute across multiple independent database nodes connected over a network, managed using the **Two-Phase Commit (2PC)**** protocol.

1. Detailed Technical Explanation

1. Two-Phase Commit (2PC) Protocol

Coordinates atomic commit/abort decisions across multiple distributed node participants.

COORDINATOR

2. Transaction Control Language (TCL) Commands

```sql

-- 1. Stop transaction

-- 2. Modify data

UPDATE Accounts SET Balance = Balance + 5000 WHERE Acc\_No = 102;

-- 3. Set a rollback checkpoint

SAVEPOINT my\_checkpoint;

UPDATE Accounts SET Balance = Balance + 5000 WHERE Acc\_No = 102;

-- 4. Rollback to Savepoint if error occurs

ROLLBACK TO my\_checkpoint;

-- 5. Commit remaining updates permanently

COMMIT;

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## 3. Database Performance Tuning Strategies

1. **Indexing Optimization:**

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**4. Quick Recall Flow**

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2PC ->  
Indexe